



Lab 1 Network Foundations and a First Design

Network System Design | SE026

Duration 120 minutes

Purpose

Build a first network proposal for a small teaching building. You will observe a connection, explain what a basic test can tell you, and link your design choices to user needs. No prior networking course is required.

What you will demonstrate

Outcome	Evidence
Identify basic device roles	Labeled endpoints, access devices and gateway
Observe a connection	Active interface, IPv4 address, mask or prefix and gateway
Interpret a diagnostic result	A ping summary and a statement of its limits
Propose a design	Requirements, topology, tradeoff and validation plan

Tools and readiness

Optional support: find the Lecture 1 PDF in this Moodle course for explanations and diagrams.

Work individually or in pairs as directed. Use paper or a drawing app and an approved connected computer. Open Windows Command Prompt or Linux/macOS Terminal; no installation or simulator is required. OS means operating system; VPN means virtual private network. Observe existing settings. Ask for help with multiple adapters, VPNs or IPv6-only connections.

Minutes	Activity
0-10	Read the brief and check tools
10-25	Identify needs and unknowns
25-45	Observe connection settings
45-65	Run and interpret ping
65-70	Short break
70-105	Draw and justify the topology
105-120	Peer review and assemble the report

1 Read the design brief

Fictional exercise: two teaching rooms serve a total of 60 student laptops. A staff office contains five fixed PCs and one network printer. Students use remote Moodle and video meetings. Staff need local printing. The budget, existing equipment, Internet capacity and cable routes have not been confirmed. The number of simultaneous users is also unknown.

For this first proposal, use one simple LAN. Treat the printer as a local service on that LAN. You may draw groups of devices if you label their counts. Do not claim your plan is a surveyed or implemented building network.

Classify planning statements

Type	Meaning
Requirement	A need that the design must satisfy
Constraint	A limit on the choices you can make
Assumption	A statement used for planning that needs confirmation
Open question	Information to ask for before committing to a design

Your planning notes

Write four requirements. Include who needs access, which service they need, and where they need it. Then write two assumptions and two open questions. Label unknown values as unknown instead of inventing a budget or speed.

Item	Your statement
Requirement 1	_____
Requirement 2	_____
Requirement 3	_____
Requirement 4	_____
Assumptions	_____ _____
Open questions	_____ _____

Checkpoint at 25 minutes: explain why one of your statements is a requirement and another is an assumption. An assumption can be reasonable while still needing confirmation.

2 Observe your current connection

Open Command Prompt on Windows or Terminal on Linux/macOS. Run the read-only commands below. Choose the active Wi-Fi or Ethernet connection rather than a disconnected adapter. Ask for help when a VPN or multiple routes complicate the choice.

OS	Commands
Windows	ipconfig If needed: route print -4
Linux	ip -br address ip route show default
macOS	route -n get default ifconfig

On Windows, match the connected adapter with its IPv4 address, subnet mask and default gateway. On Linux, the default route lists via (gateway) and dev (interface). On macOS, read the gateway and interface from the route output, then find that interface in ifconfig. Do not assume the interface is always en0.

A worked reading example

Sample only: interface Wi-Fi, address 192.168.10.25, mask 255.255.255.0, gateway 192.168.10.1. The address belongs to the interface. The gateway is the next router for the default route. This mask is also written /24. You do not need subnet calculations today, and you must not copy these settings into your computer.

Your observation record

Field	Observed value
OS and connection type	_____
Active interface	_____
IPv4 address	_____
Mask or prefix	_____
Default gateway	_____
Test time and VPN status	_____

Capture one relevant screenshot or copy the output. Caption it with the interface you chose and how you identified the gateway. Record unavailable values honestly. If the network has no usable IPv4 settings, analyze the sample above and label it as sample evidence.

Checkpoint at 45 minutes: show the interface and gateway evidence. Your personal or classroom network is an observation, not proof of the fictional building topology.

3 Run and interpret a local ping

Ping sends ICMP echo probes and waits for replies. ICMP means Internet Control Message Protocol. A reply supports IP reachability for that probe. Round-trip time (RTT) measures elapsed time from sending it to receiving the reply, in milliseconds. It includes the outward and return paths.

First ping loopback, 127.0.0.1. It tests the local IPv4 stack inside your computer. Next ping the gateway address you recorded, if available. Replace TARGET with that address. For example, if your own gateway is 192.168.10.1, use that value. Do not type TARGET literally.

OS	Four-probe commands
Windows	ping -4 -n 4 127.0.0.1 ping -4 -n 4 TARGET
Linux	ping -4 -c 4 127.0.0.1 ping -4 -c 4 TARGET
macOS	ping -c 4 127.0.0.1 ping -c 4 TARGET (use the IPv4 address)

Record the summary. If no reply arrives, write RTT unavailable. A displayed time below 1 ms is not proof of zero delay. A timeout can reflect filtering or a connectivity problem. Preserve the failed result and discuss it with the teacher.

Target	Sent and replies	RTT summary or unavailable
Loopback	____ sent / ____ replies	_____
Your gateway	____ sent / ____ replies	_____

Interpretation

1. Explain what loopback tests and what it does not test.
2. Does a gateway reply prove that a remote Moodle login works? Explain.
3. State one possible reason for a missing gateway reply.
4. Explain why ping alone cannot choose the Internet capacity needed by 60 users.

Ping measures probe response time, not application traffic demand or available download throughput. Capacity planning also needs expected simultaneous users and application demand. An unanswered test with a reasonable explanation is acceptable evidence. If tests are unavailable, use the labeled sample: four sent, three replies, RTTs 12, 15 and 18 ms, average 15 ms.

Checkpoint at 65 minutes: show the summary and one interpretation. Optional: test an instructor-approved remote IPv4 address after the core tests.

4 Draw your first network proposal

Return to the fictional building brief. Draw a topology showing the two teaching rooms, staff office, student laptops, office PCs, printer, access devices, a gateway router, ISP/Internet connection and remote Moodle service. A clearly labeled hand sketch is sufficient.

An access point (AP) connects Wi-Fi devices to the LAN. A switch connects Ethernet devices within the LAN. A gateway router connects it to other networks. ISP means Internet service provider. Choose wired, wireless or mixed access for each group. Label roles and use an Ethernet/Wi-Fi legend. An icon may represent a group of devices, but show the group count. Label any AP count or placement as provisional until coverage and simultaneous demand are checked.

Explain the paths

Write the devices and links used by a student laptop to reach the local printer. Then write its path toward remote Moodle. Explain which traffic can remain within the simple LAN and which needs the gateway and ISP.

Justify your choices

User need	Proposed choice	Benefit and limitation
_____	_____	_____ _____
_____	_____	_____ _____

Choose one assumption that would change your design if it proved wrong. Explain how you would check it. Then identify one shared device or link whose failure would affect multiple users.

Plan validation

For two requirements, describe a test and the result you would consider acceptable. For example, a requirement about using Moodle needs an application test, not just a ping reply. Propose checks for the fictional building. Do not run load tests or change shared infrastructure during this lab.

Peer review

Ask a partner to follow both service paths on your drawing. Can they identify every device role, link type and unknown? Revise unclear labels. Check that your plan supports both mobile laptops and fixed office devices.

Checkpoint at 105 minutes: show your topology, four requirements, two assumptions, two open questions, one tradeoff, one failure point and two validation ideas. The exact equipment brand and IP addressing plan are not required.

5 Assemble your report

Prepare one PDF named SE026_Lab1_StudentID.pdf. If working in a pair, identify both students and follow the teacher's submission directions. The teacher will specify the due date and submission location.

Section	Include
Planning	Four requirements, two assumptions and two open questions
Connection evidence	OS, time, interface, address, mask or prefix, gateway and captioned output
Ping interpretation	Summary, four short explanations and any test limitations
Design proposal	Labeled topology, local and remote paths, justified access choice
Validation	One failure point and checks for two requirements

Task rubric

Criterion	Points
Requirements and clear distinction between assumptions and unknowns	2
Coherent topology with correct device roles and links	3
Connection and ping evidence with sound interpretation	2
Tradeoff, failure point and validation reasoning	2
Clear report and honestly labeled evidence	1
Total	10

The rubric scores this task out of 10. Label copied samples as samples. Distinguish your own observations from the fictional proposal. More screenshots do not replace explanations.

Optional extension after the core work

If time remains, ask the teacher for a guided traceroute demonstration or a transmission-delay exercise. These extend the lesson and are not required for this report. Complete the core topology and explanations first.

Next lab

Use protocol-layer concepts and a prepared packet trace to understand how services communicate. The design proposal can then develop with more evidence. Simulator setup and device configuration will follow when needed.